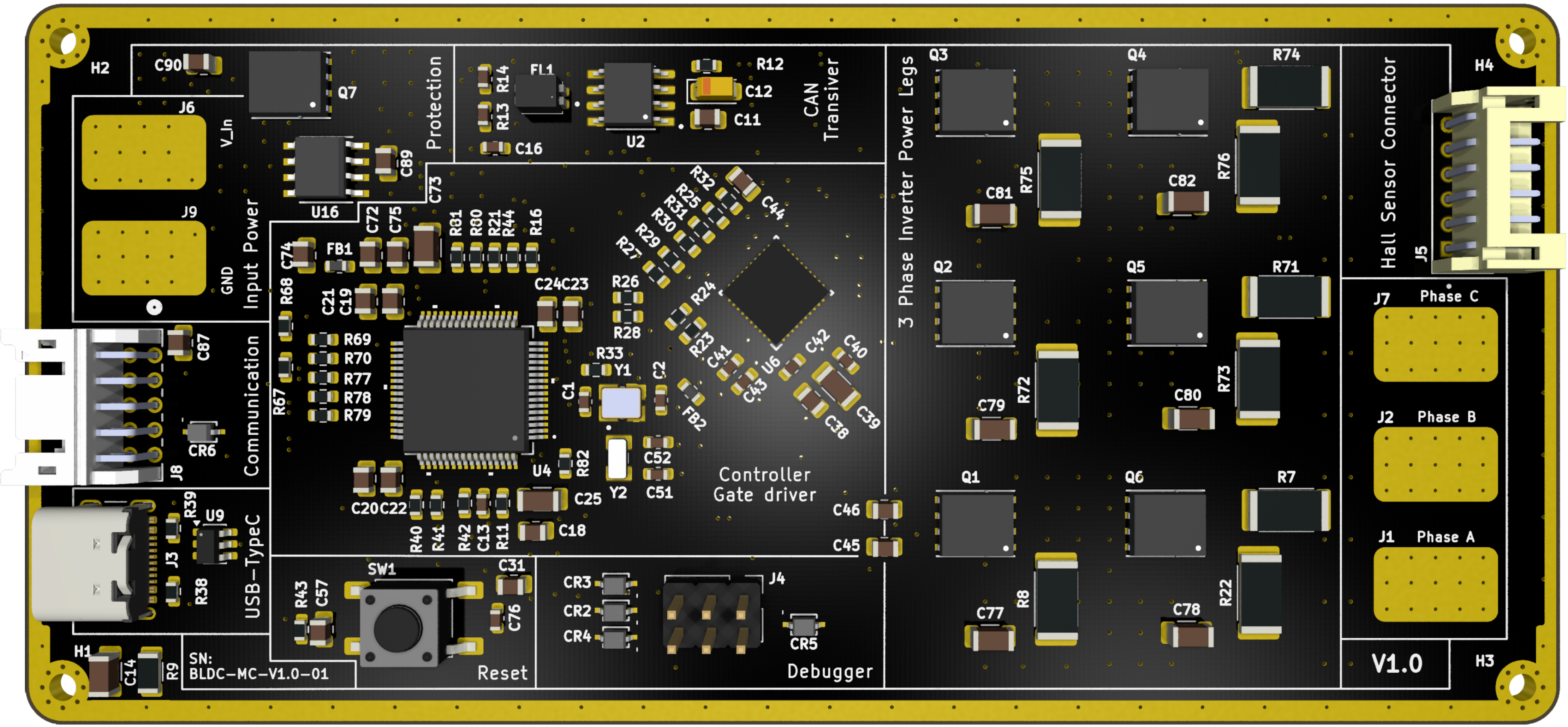
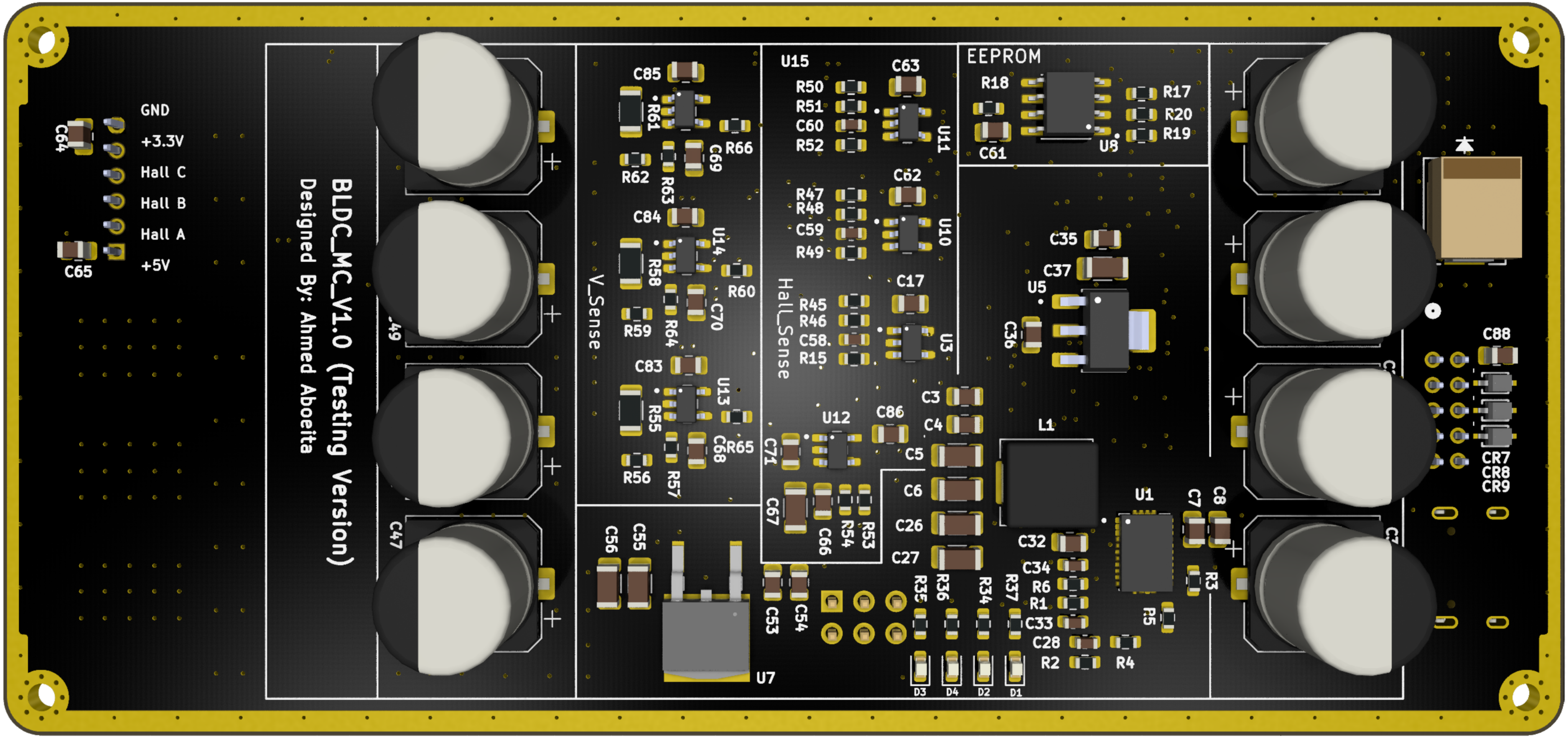
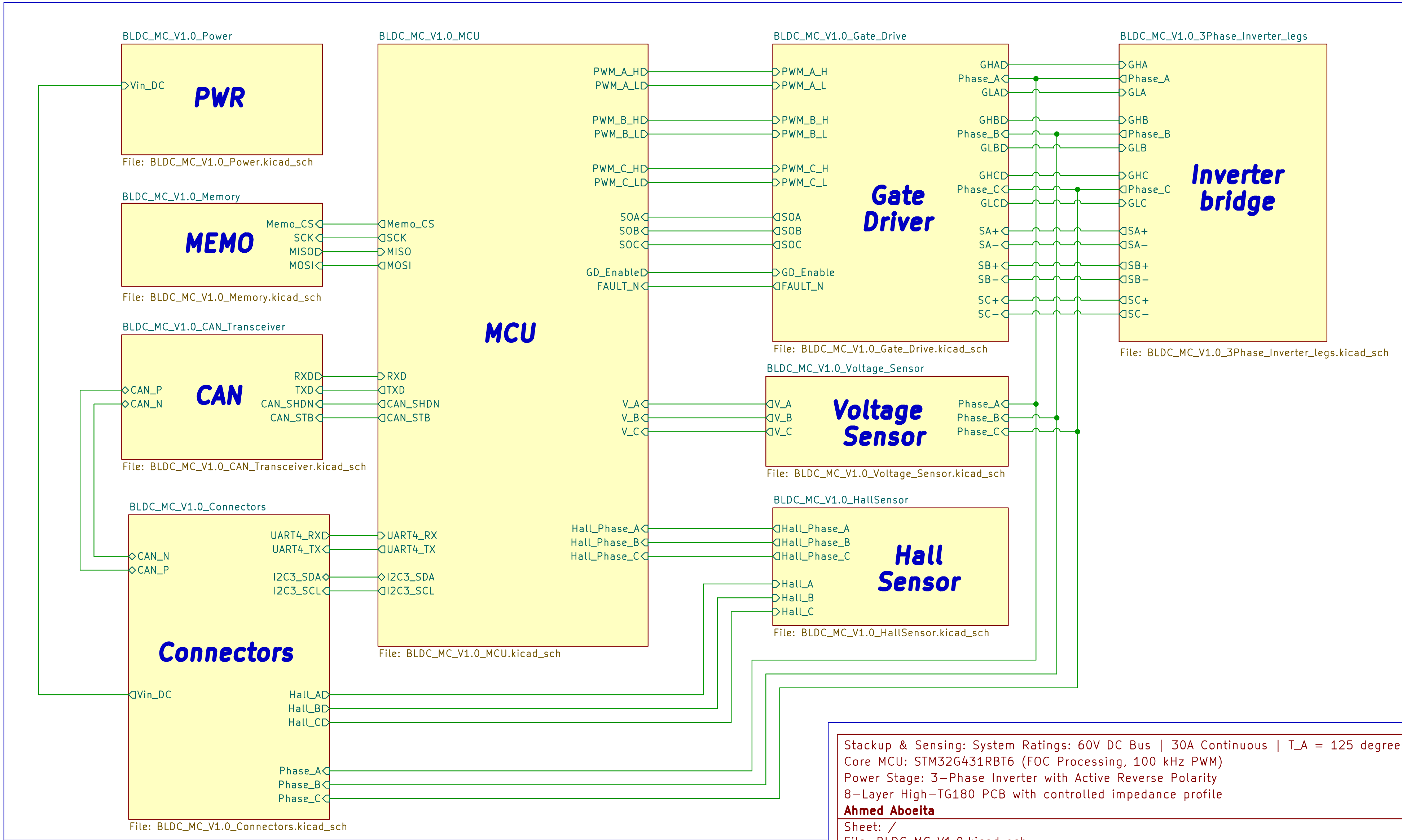

Protection
CAN Transiver
3 Phase Inverter Power Legs
Hall Sensor Connector
V_In
GND
Input Power
Communication
USB-TypeC
Controller
Gate driver
Reset
Debugger
SN:
BLDC-MC-V1.0-01
V1.0
Phase C
Phase B
Phase A


BLDC_MC_V1.0 (Testing Version)
Designed By: Ahmed Aboeita
GND
+3.3V
Hall C
Hall B
Hall A
+5V
V_Sense
Hall_Sense
EEPROM

[1] BLDC_MC V1 60V 30A



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3−Phase Inverter with Active Reverse Polarity
8−Layer High−TG180 PCB with controlled impedance profile

Ahmed Aboeita

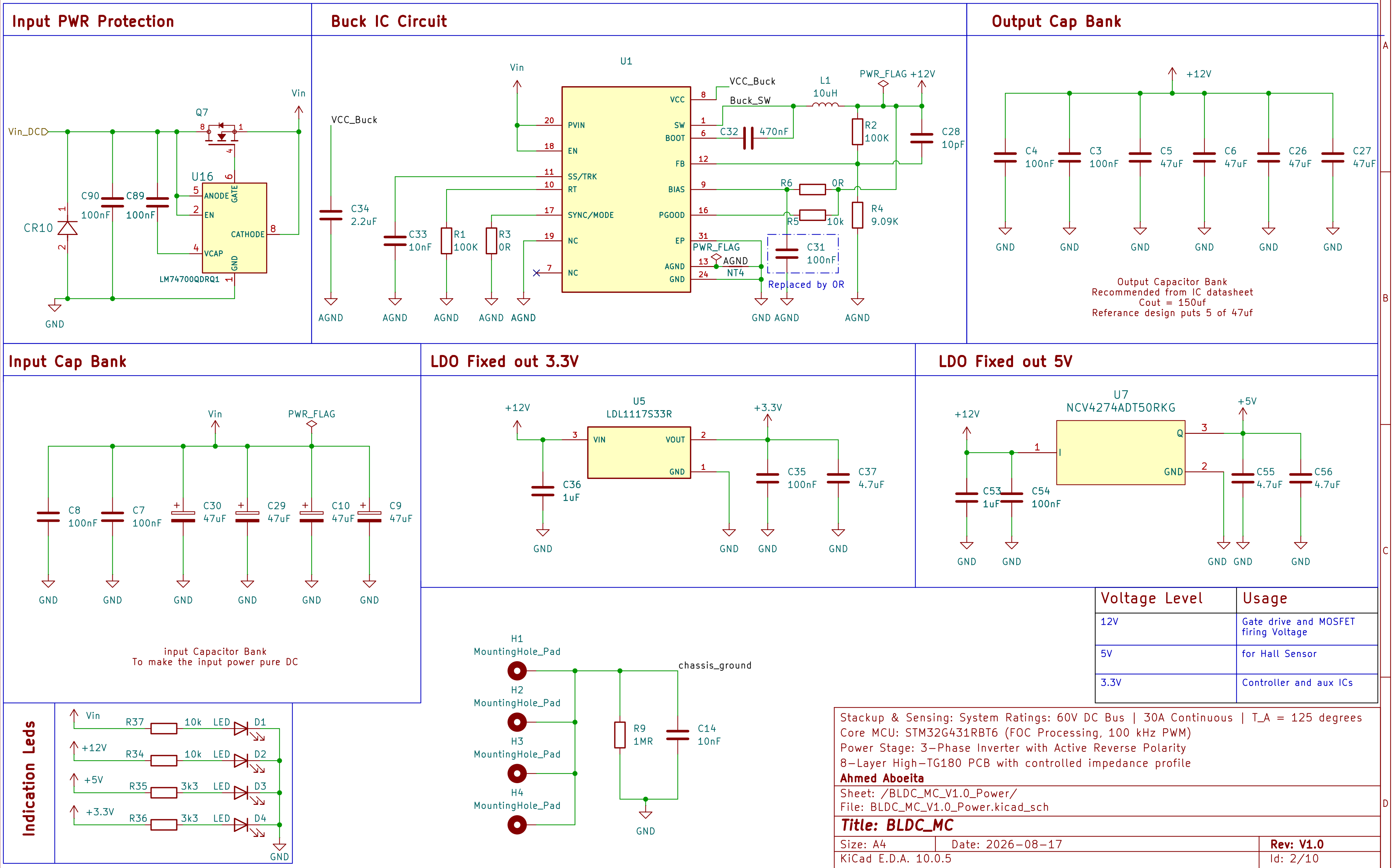
Sheet: /
File: BLDC_MC_V1.0.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026−08−17	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 1/10

[2] Power Conversion Vin to 12V and 5V to 3.3V

The maximum input votage of this 60V so we can use for BLDC motors up to 60V



Voltage Level	Usage
12V	Gate drive and MOSFET firing Voltage
5V	for Hall Sensor
3.3V	Controller and aux ICs

Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity
8-Layer High-TG180 PCB with controlled impedance profile
Ahmed Aboeita

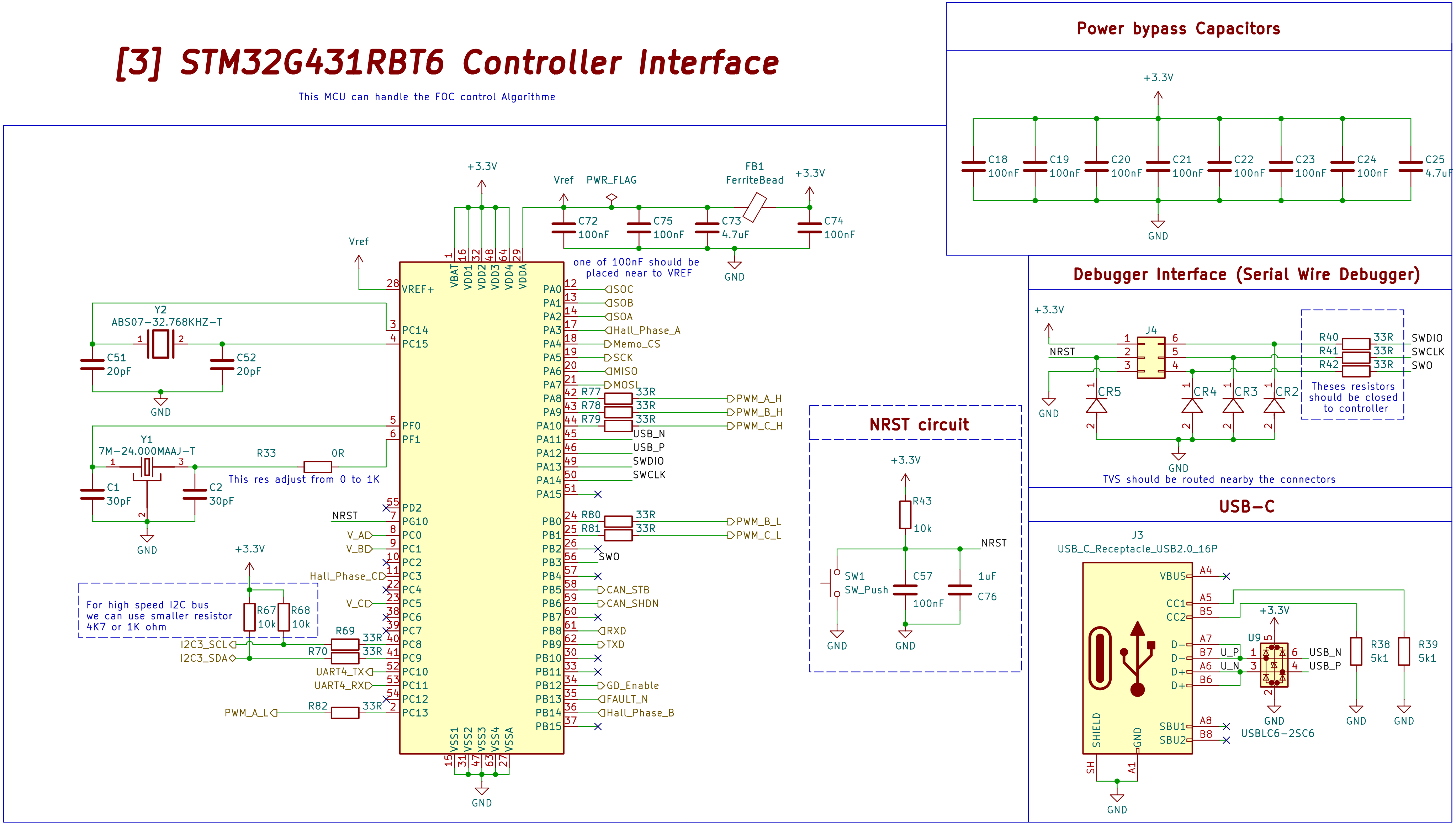
Sheet: /BLDC_MC_V1.0_Power/
File: BLDC_MC_V1.0_Power.kicad_sch

Title: BLDC_MC

Size: A4 | Date: 2026-08-17 | **Rev: V1.0**
KiCad E.D.A. 10.0.5 | Id: 2/10

[3] STM32G431RBT6 Controller Interface

This MCU can handle the FOC control Algorithme



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity
8-Layer High-TG180 PCB with controlled impedance profile
Ahmed Aboeita

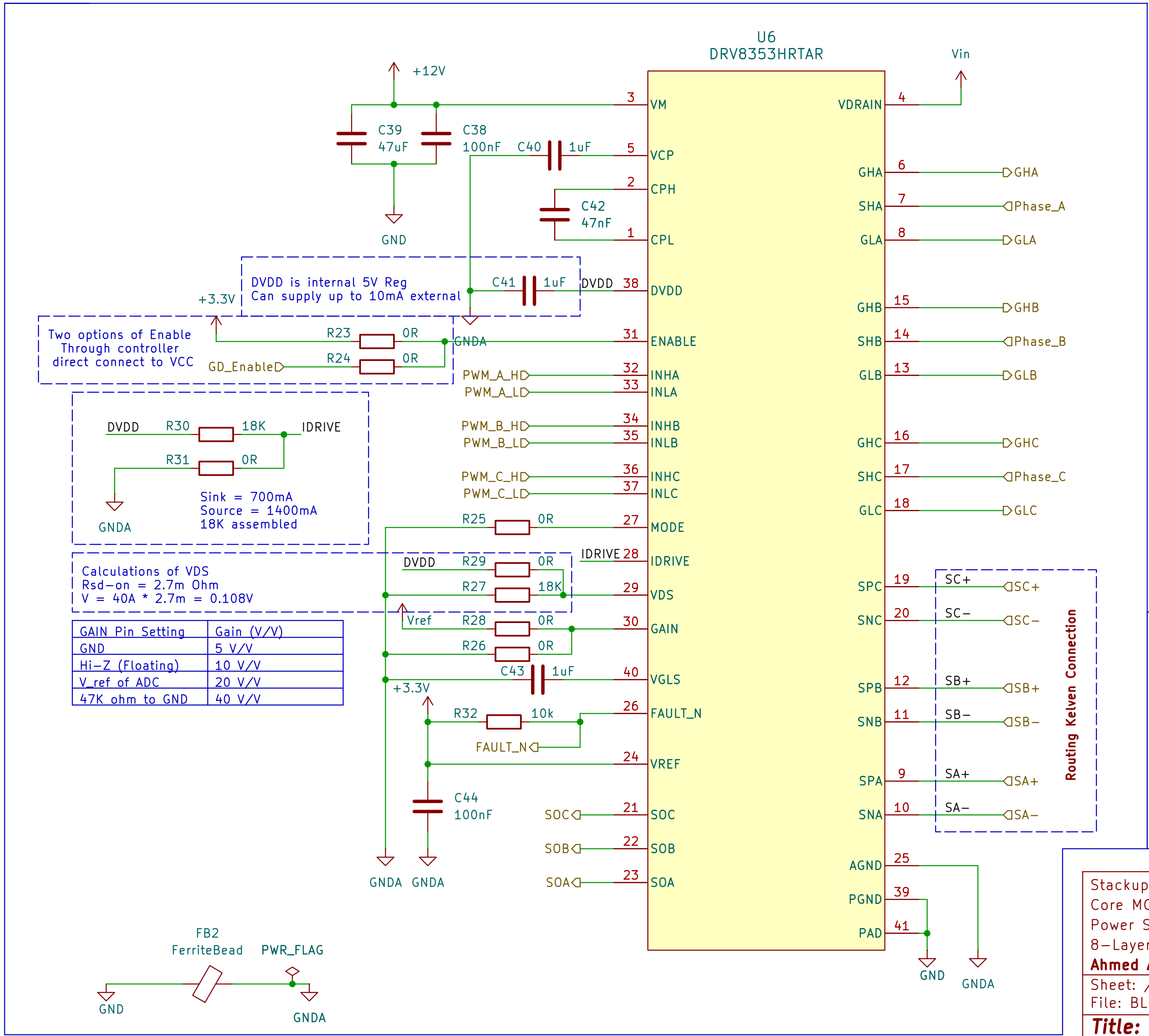
Sheet: /BLDC_MC_V1.0_MCU/
File: BLDC_MC_V1.0_MCU.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-08-17	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 3/10

[4] Smart Gate Drive

for the 3 phase inverter with current sensor



GAIN Pin Setting	Gain (V/V)
GND	5 V/V
Hi−Z (Floating)	10 V/V
V_ref of ADC	20 V/V
47K ohm to GND	40 V/V

Gate Resistors vs. IDRIVENormal Driver:

Uses external gate resistors to limit the current charging the MOSFET gate.
If you want to change the switching speed to reduce EMI or heat, you have to desolder and change the resistors.

Smart Driver: Uses an internal, programmable current source called IDRIVE. You set the source/sink current (e.g., 100mA, 1A) via a single pin or SPI. This allows you to tune the switching speed of your inverter without changing any hardware components.

Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3−Phase Inverter with Active Reverse Polarity
8−Layer High−TG180 PCB with controlled impedance profile

Ahmed Aboeita

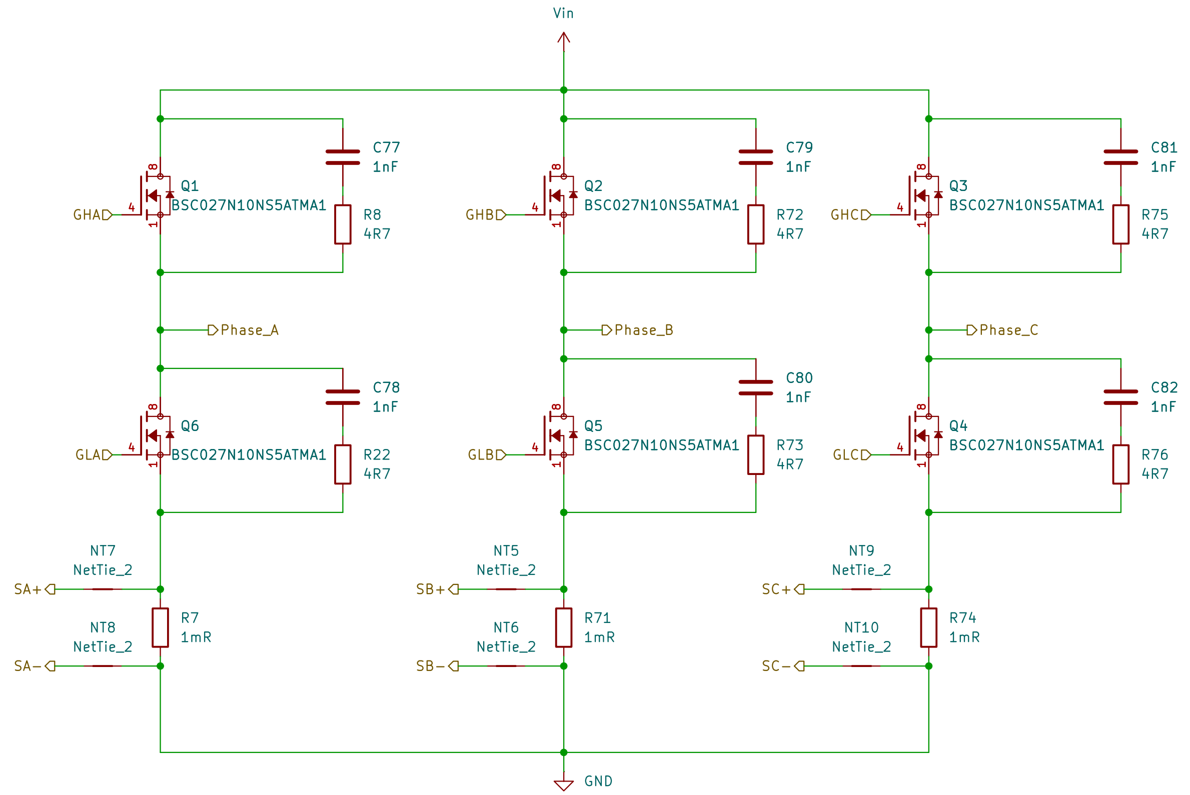
Sheet: /BLDC_MC_V1.0_Gate_Drive/
File: BLDC_MC_V1.0_Gate_Drive.kicad_sch

Title: BLDC_MC

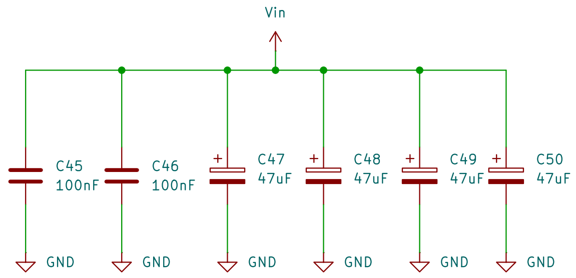
Size: A4	Date: 2026−08−17	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 4/10

[5] 3 Phases Inverter Bridge

3 Phase Inverter Bridge



DC Link



Steps for adjusting snubber circuit

1− Measure the Initial Ringing
2− Add the "Shift" Capacitor −
increase the value of C_add until the ringing
frequency is exactly half of the original frequency f_new = 0.5 * f_ring).
3− Calculte the parasitic capacitance C_p = C_add / 3
4− Calculate Parasitic Inductance L_p −> L_p = 1/((2*pi*f_ring)^2 * C_p)
5− Determine the Snubber Resistor R_snub −> R_snub = sqrt{L_p/C_p}

Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3−Phase Inverter with Active Reverse Polarity
8−Layer High−TG180 PCB with controlled impedance profile

Ahmed Aboeita

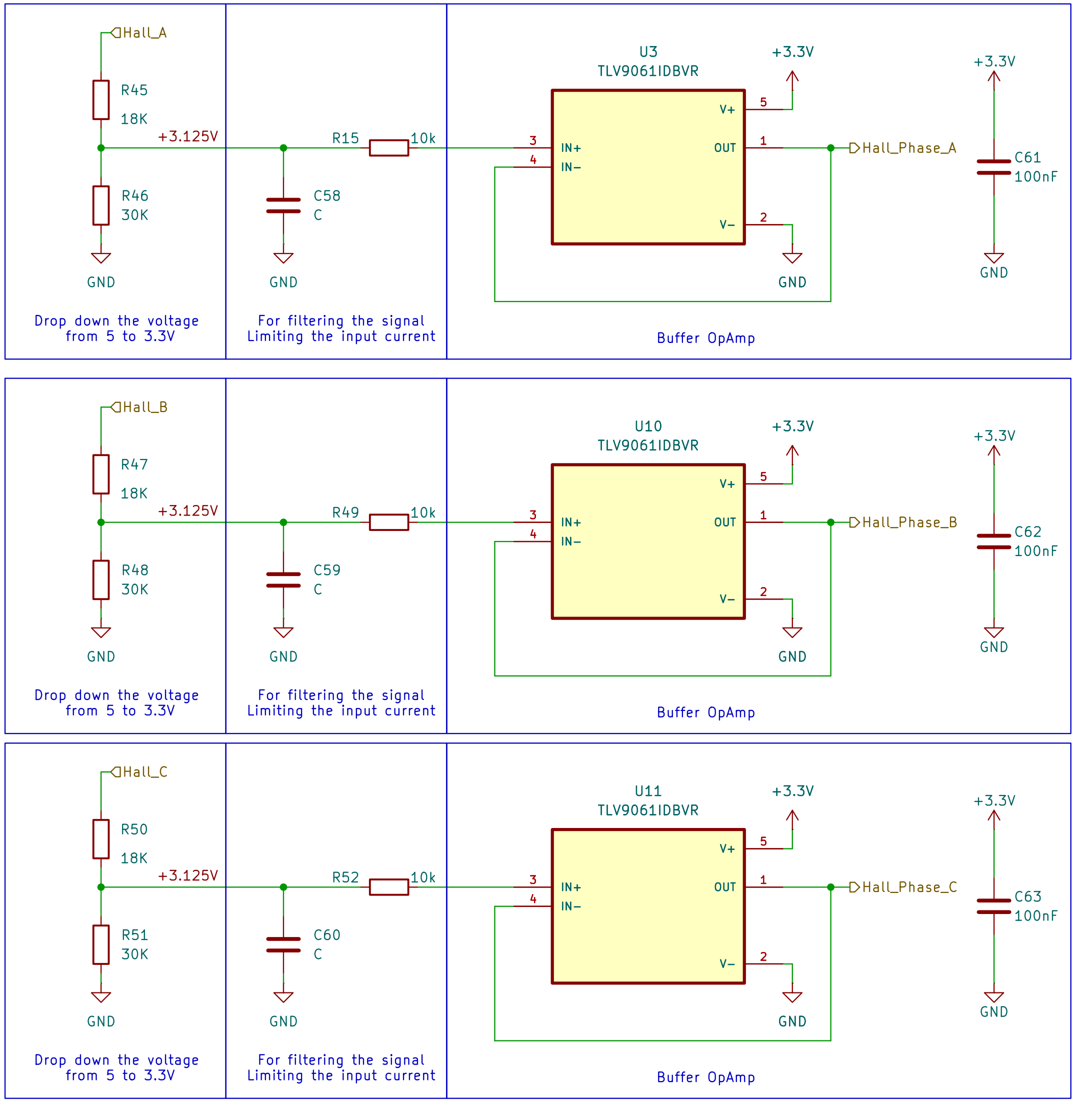
Sheet: /BLDC_MC_V1.0_3Phase_Inverter_legs/
File: BLDC_MC_V1.0_3Phase_Inverter_legs.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026−08−17	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 5/10

[6] Hall Effect Sensor Interface

For the 3 Phases of Hall Sensor



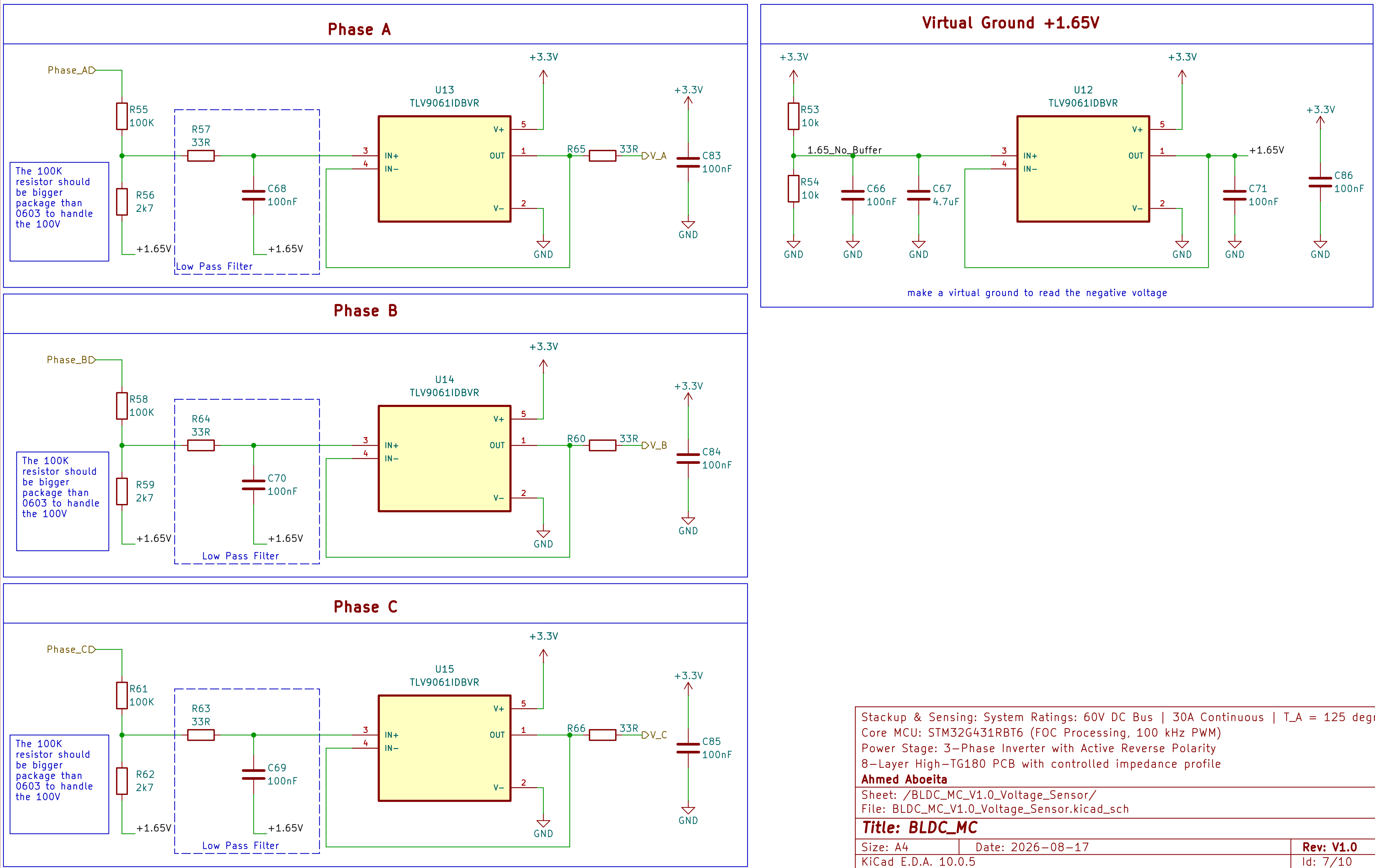
Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity
8-Layer High-TG180 PCB with controlled impedance profile
Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_HallSensor/
File: BLDC_MC_V1.0_HallSensor.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-08-17	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 6/10

[7] 3 Phase Inverter Voltage Sensing



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity
8-Layer High-TG180 PCB with controlled impedance profile

Ahmed Aboeita

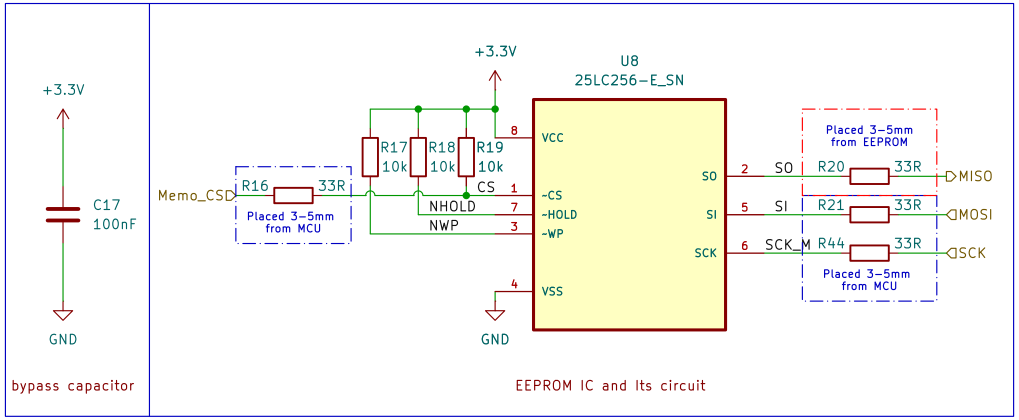
Sheet: /BLDC_MC_V1.0_Voltage_Sensor/
File: BLDC_MC_V1.0_Voltage_Sensor.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-08-17	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 7/10

[8] EEPROM Memory 256Kbit

This memory used to store important data and configration parameters



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity
8-Layer High-TG180 PCB with controlled impedance profile
Ahmed Aboeita

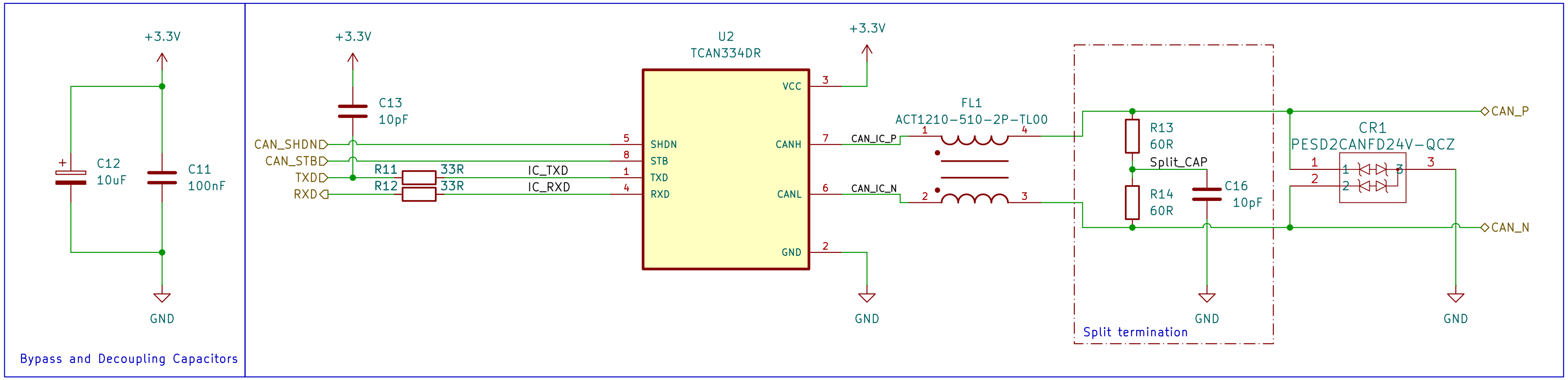
Sheet: /BLDC_MC_V1.0_Memory/
File: BLDC_MC_V1.0_Memory.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-08-17	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 8/10

[9] CAN Transiver Classic CAN 1Mb/S

This IC can operate as CAN−FD for faster datarate



Note

Series Resistors Help manage signal integrity
by damping potential high−frequency noise or reflections
on the digital traces.

Parallel Capacitors is used to filter noise on the digital I/O lines

Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3−Phase Inverter with Active Reverse Polarity
8−Layer High−TG180 PCB with controlled impedance profile

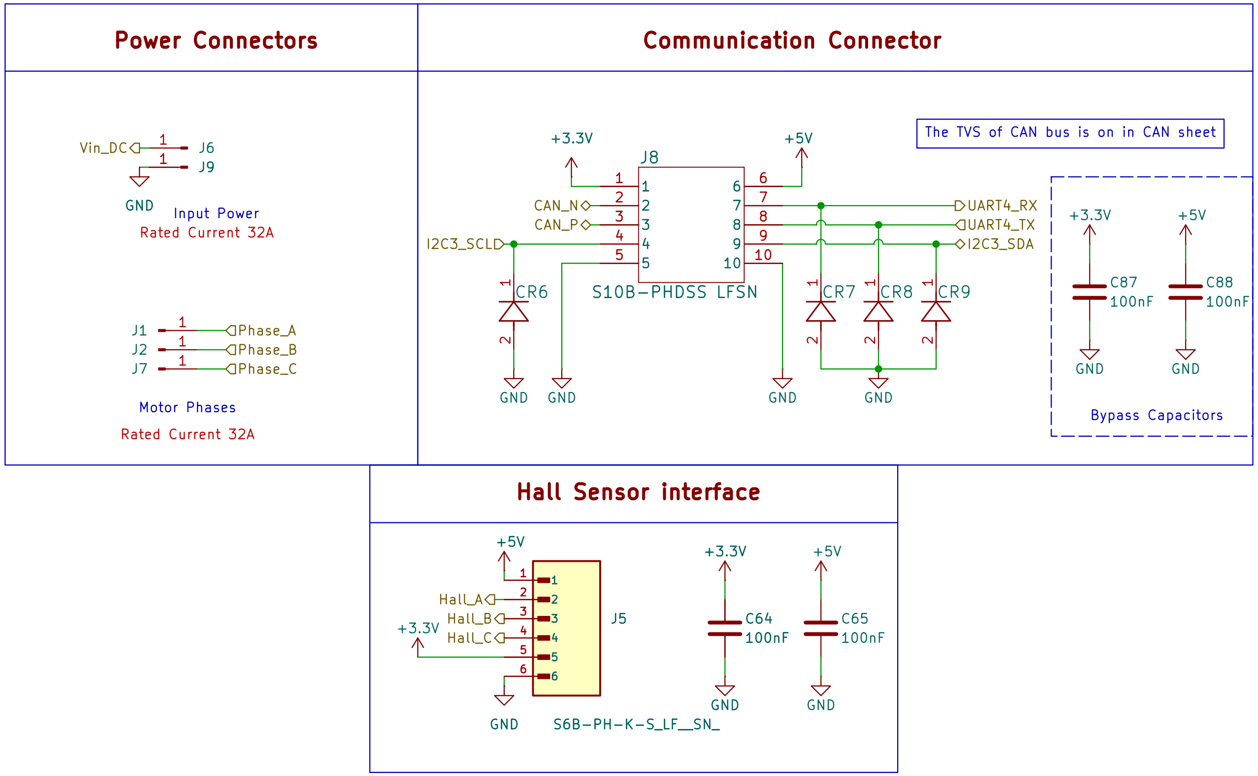
Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_CAN_Transceiver/
File: BLDC_MC_V1.0_CAN_Transceiver.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026−08−17	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 9/10

[10] Board's Connectors Power & Communication

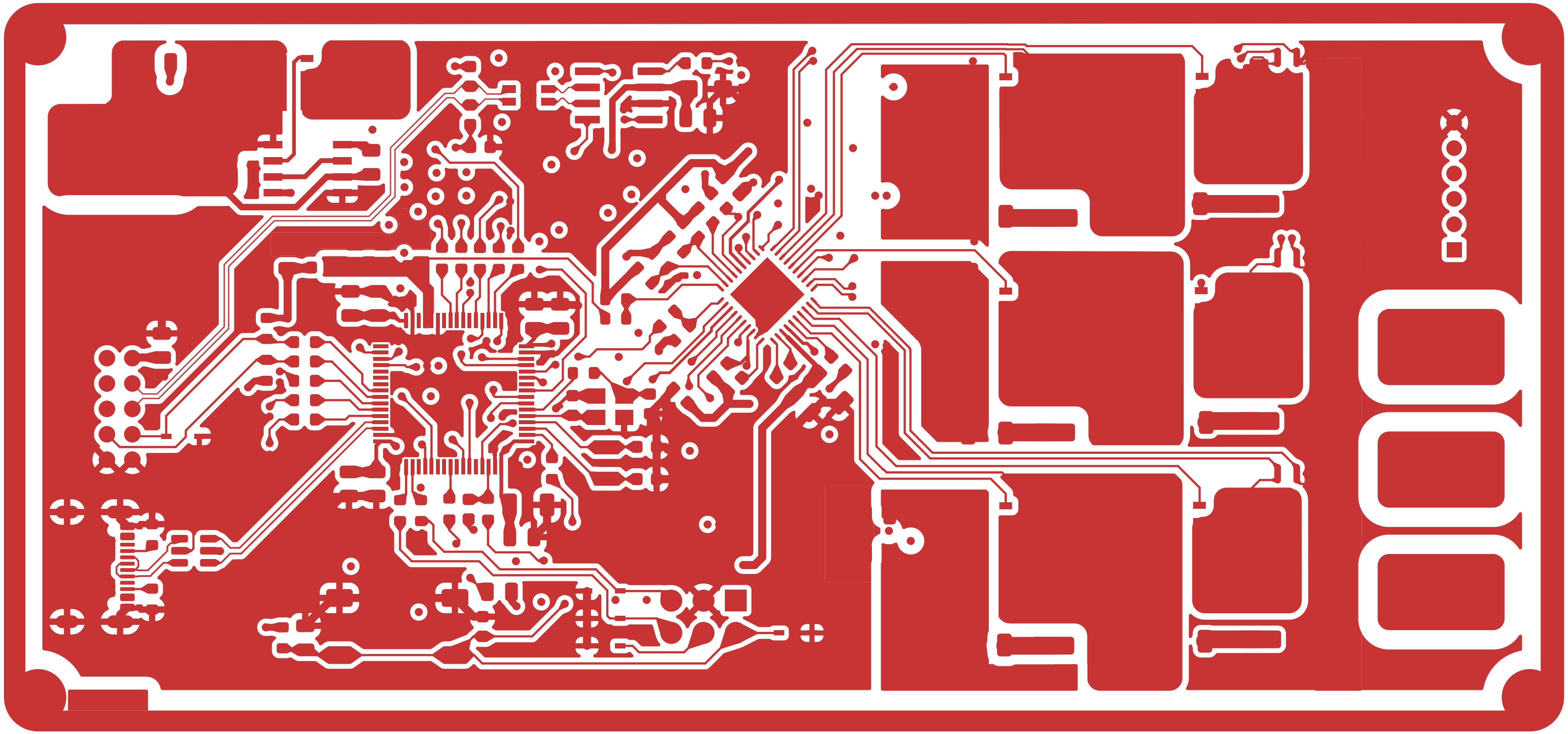


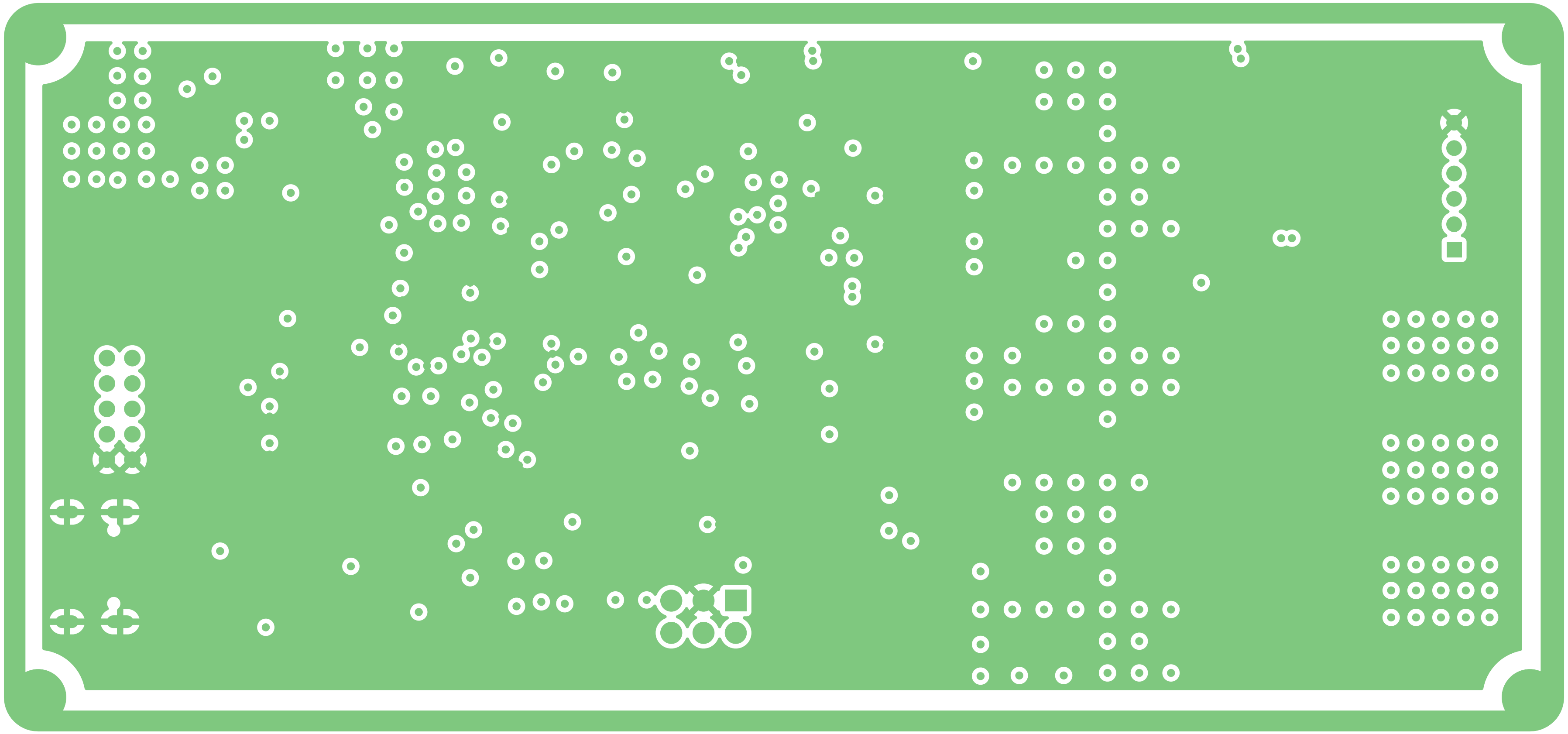
Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity
8-Layer High-TG180 PCB with controlled impedance profile
Ahmed Aboeita

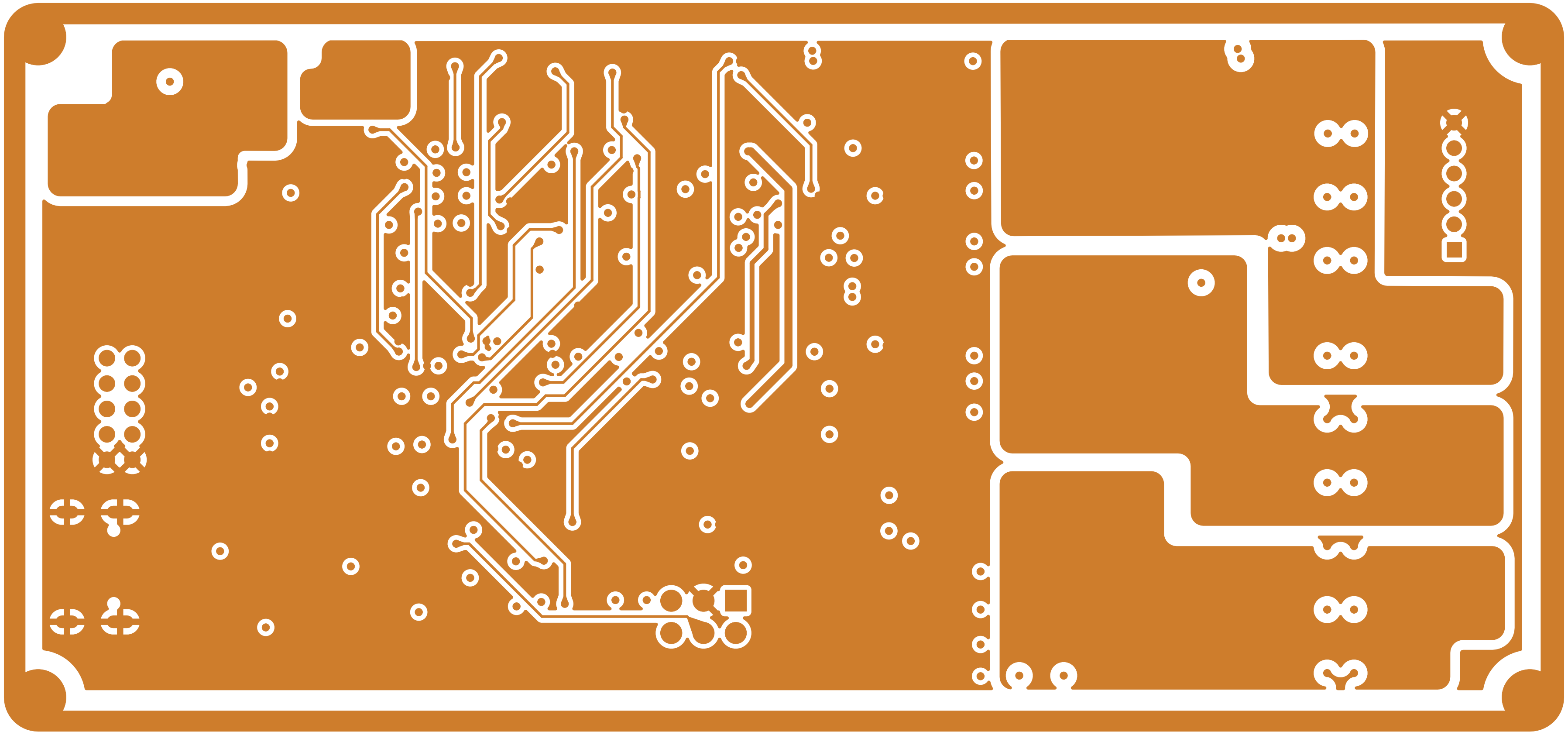
Sheet: /BLDC_MC_V1.0_Connectors/
File: BLDC_MC_V1.0_Connectors.kicad_sch

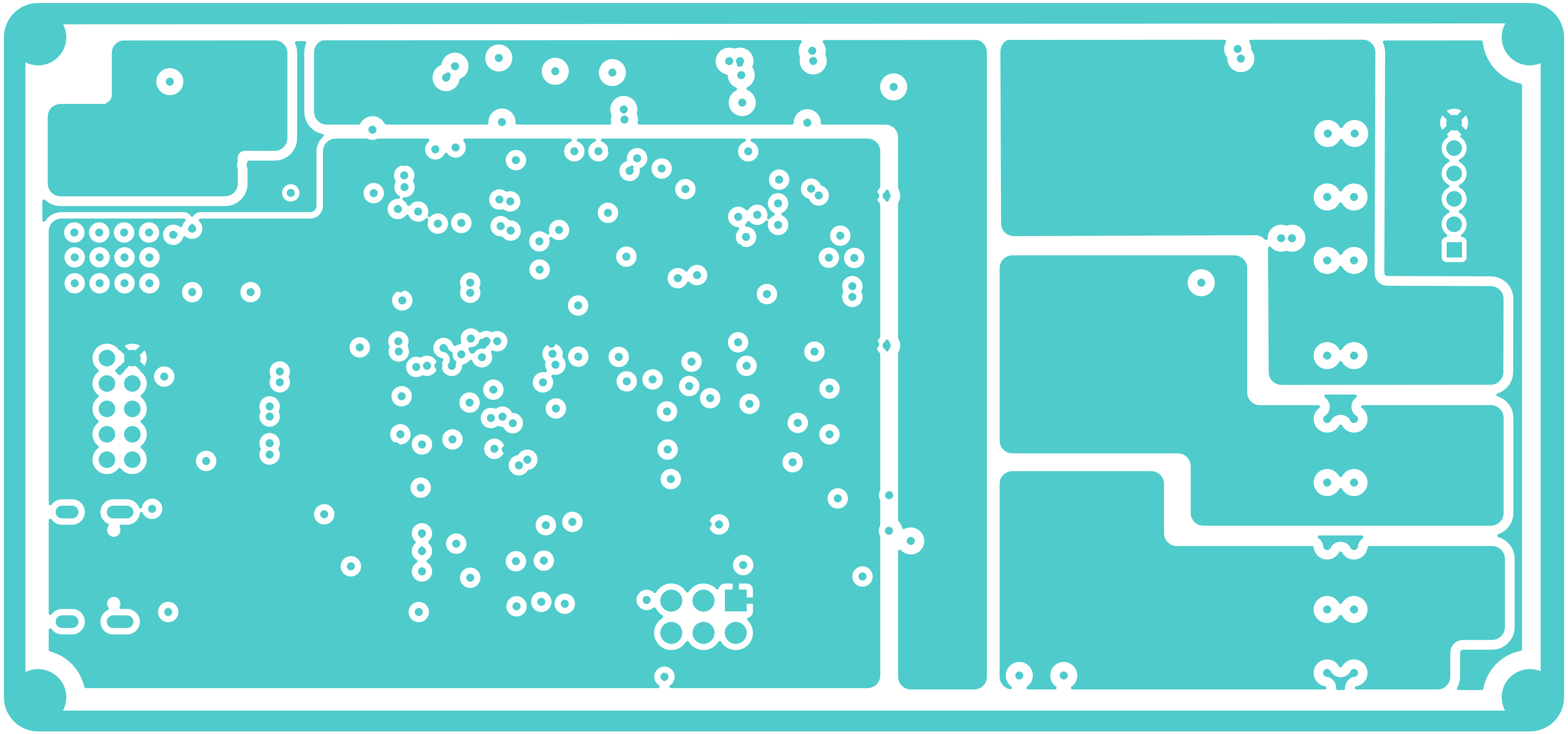
Title: BLDC_MC

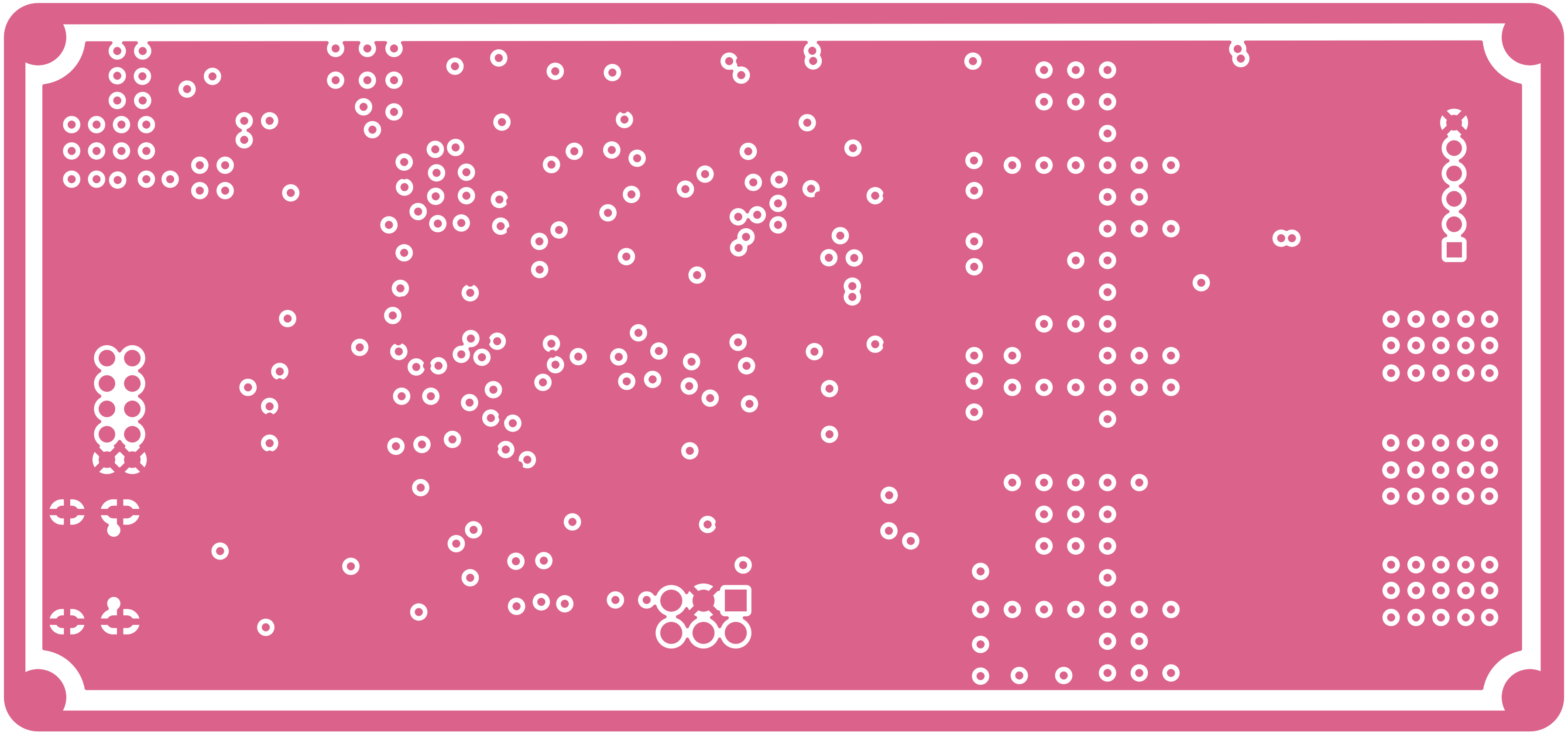
Size: A4	Date: 2026-08-17	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 10/10

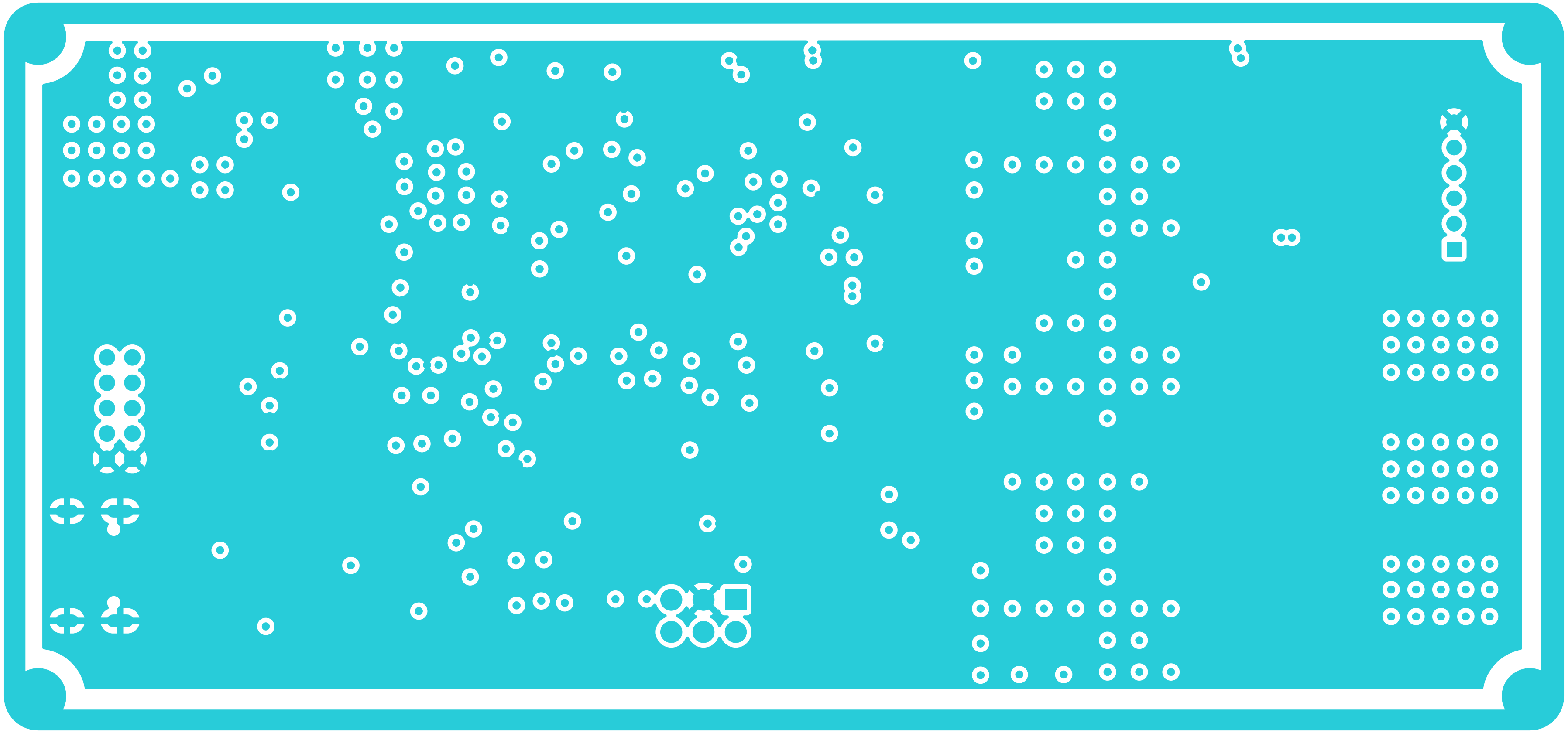


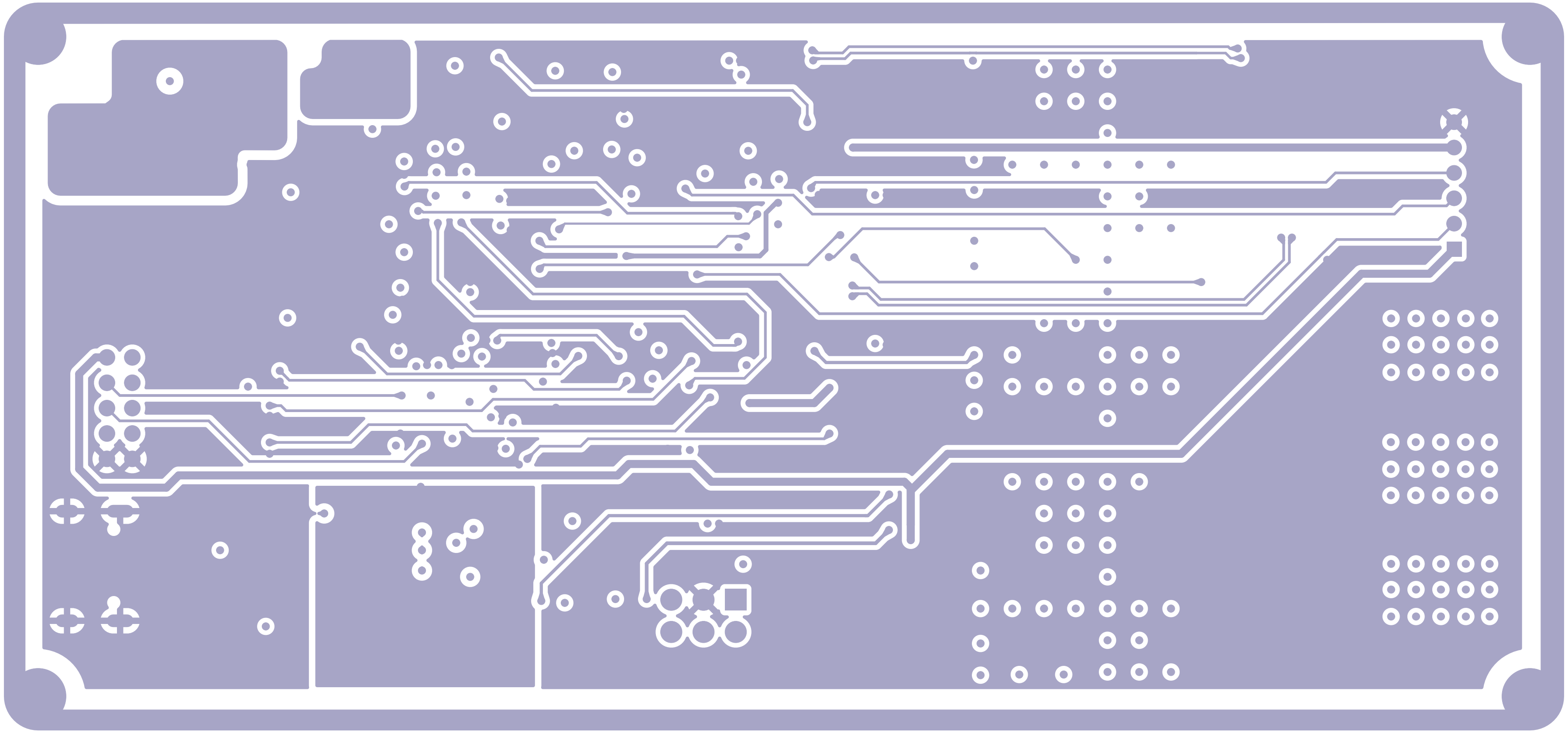


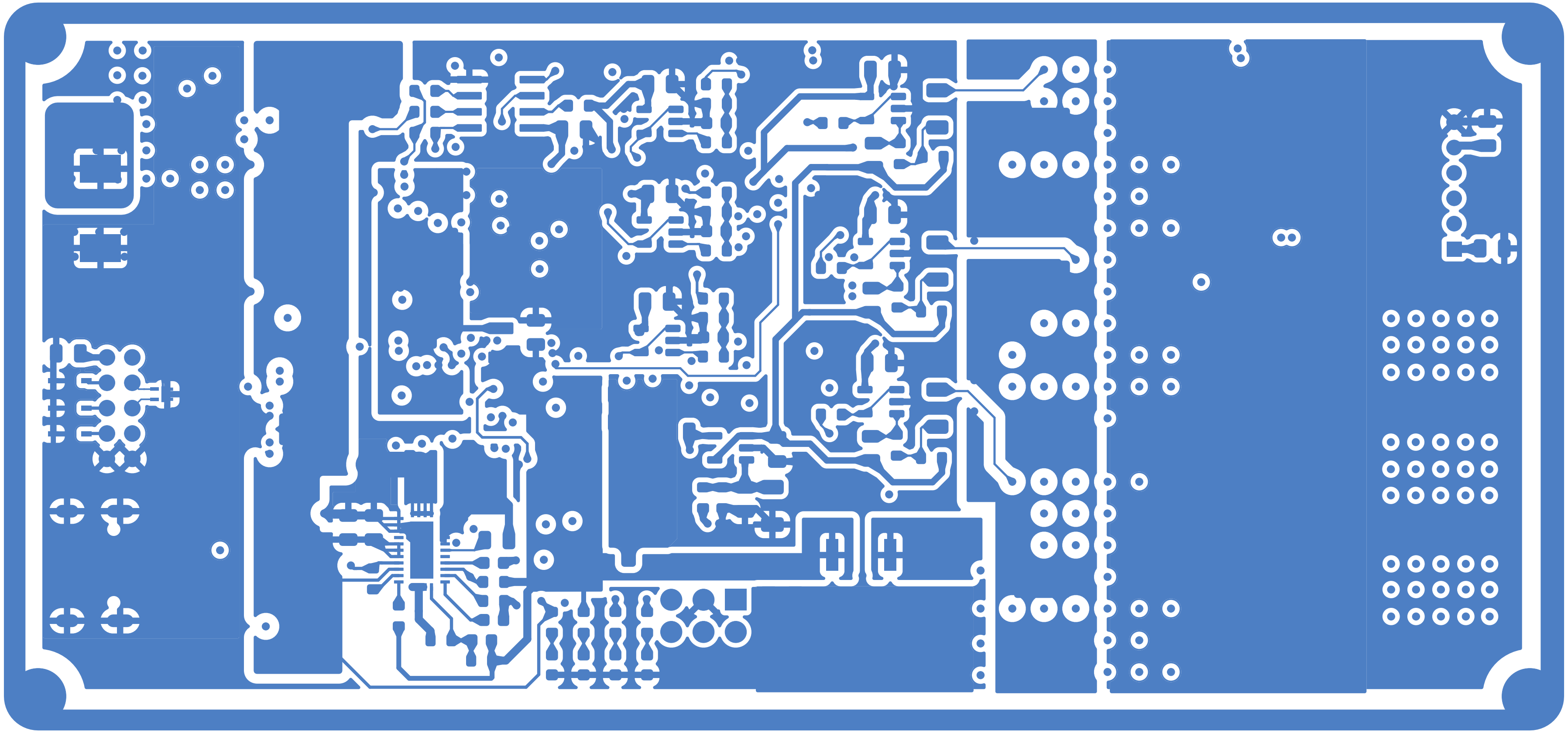












MP	Qty	Reference	Value	Assembled	LCSC_PN	Description	Footprint
CC0603JRNPO9BN300	2	C1,C2	30pF	Yes	C113797	Unpolarized capacitor	Capacitor_SMD:C_0603_1608Metric
CL21B104KCFNNNE	42	C3,C4,C7,C8,C11,C17,C18,C19,C20,C21,C22,C23,C24,C31,C35,C38,C44,C45,C46,C54,C57,C61,C62,C63,C64,C65,C66,C68,C69,C70,C71,C72,C74,C75,C83,C84,C85,C86,C87,C88,C89,C90	100nF	No,Yes	C28233	Unpolarized capacitor	Capacitor_SMD:C_0805_2012Metric
GRM31CR61E476ME44L	5	C5,C6,C26,C27,C39	47uF	Yes	C403725	Unpolarized capacitor	Capacitor_SMD:C_1206_3216Metric
118EC449	8	C9,C10,C29,C30,C47,C48,C49,C50	47uF	Yes	C46528101	Polarized capacitor	Capacitor_SMD:CP_Elec_10x10.5
TAJA106K016RNJ	1	C12	10uF	Yes	C7171	Polarized capacitor	Capacitor_Tantalum_SMD:CP_EIA-3216-18_Kemet-A
CL10C100JB8NNNC	3	C13,C16,C28	10pF	No	C1634	Unpolarized capacitor	Capacitor_SMD:C_0603_1608Metric
CL31B475KBHNNNE	6	C25,C37,C55,C56,C67,C73	4.7uF	Yes	C51205	Unpolarized capacitor	Capacitor_SMD:C_1206_3216Metric
CL21B474KBFNNNE	1	C32	470nF	Yes	C13967	Unpolarized capacitor	Capacitor_SMD:C_0805_2012Metric
CC0603KRX7R9BB103	1	C33	10nF	Yes	C100042	Unpolarized capacitor	Capacitor_SMD:C_0603_1608Metric
GRM188Z71E225KE43D	1	C34	2.2uF	Yes	C415535	Unpolarized capacitor	Capacitor_SMD:C_0603_1608Metric
CL21B105KBFNNNE	2	C36,C53	1uF	Yes	C28323	Unpolarized capacitor	Capacitor_SMD:C_0805_2012Metric
CL10B105KA8NNNC	4	C40,C41,C43,C76	1uF	Yes	C29936	Unpolarized capacitor	Capacitor_SMD:C_0603_1608Metric
CL10B473KB8NNNC	1	C42	47nF		C1622	Unpolarized capacitor	Capacitor_SMD:C_0603_1608Metric
CL10C200JB8NNNC	2	C51,C52	20pF	Yes	C1648	Unpolarized capacitor	Capacitor_SMD:C_0603_1608Metric
CC1206KKX7RDBB102	6	C77,C78,C79,C80,C81,C82	1nF	No	C23631	Unpolarized capacitor	Capacitor_SMD:C_1206_3216Metric
PESD2CANFD24V-QCZ	1	CR1	PESD2CANFD24V-QCZ	Yes	C6251381	42VC Clamp 2.6A@8/20us Ipp ESD DIODE DFN1412D-3	BLDC_Motor_Driver_Components_Footprint:DFN1412D-3_SOT8009_NEX
PESD3V3L1BAZ	8	CR2,CR3,CR4,CR5,CR6,CR7,CR8,CR9	PESD3V3L1BA_115	Yes	C552513	26VC Clamp ESD DIODE SOD-323	BLDC_Motor_Driver_Components_Footprint:DIODE_PESD3V3L1BA_115_NEX
SMDJ110CA	1	CR10	SMDJ110CA		C2990268	177VC Clamp 16.9A Ipp TVS DIODE SMC(DO-214AB)	BLDC_Motor_Driver_Components_Footprint:SMDJSeries_LTF
YLED0603R	4	D1,D2,D3,D4	LED	Yes	C19171390	Light emitting diode	LED_SMD:LED_0603_1608Metric
BLM18AG601SN1D	2	FB1,FB2	FerriteBead	Yes	C19330	Ferrite bead	Inductor_SMD:L_0603_1608Metric
ACT1210-510-2P-TL00	1	FL1	ACT1210-510-2P-TL00	Yes	C95572		BLDC_Motor_Driver_Components_Footprint:FIL_ACT1210-510-2P-TL00
TYPE-C 16PIN 2MD(073)	1	J3	USB_C_Receptacle_USB2.0_16P	Yes	C2765186	USB 2.0-only 16P Type-C Receptacle connector	BLDC_Motor_Driver_Components_Footprint:USB_C_Receptacle_GCT_USB4105-xx-A_16P_TopMnt_Horizontal
S6B-PH-K-S(LF)(SN)	1	J5	S6B-PH-K-S_LF__SN_	Yes	C157920	Connector Header Through Hole, Right Angle 6 position 0.079 (2.00mm)	BLDC_Motor_Driver_Components_Footprint:JST_S6B-PH-K-S_LF__SN_
S10B-PHDSS(LF)(SN)	1	J8	S10B-PHDSS LFSN	Yes	C161672	CONN HEADER TH R/A 10POS 2mm	BLDC_Motor_Driver_Components_Footprint:CONN_S10B-PHDSS_JST
SMMS0650-100M	1	L1	10uH	Yes	C149584	Inductor	Inductor_SMD:L_KOHERelec_MDA7030
BSC027N10NS5ATMA1	7	Q1,Q2,Q3,Q4,Q5,Q6,Q7	BSC027N10NS5ATMA1		C534315	N-Channel 100V 3W Surface Mount TDSON-8-EP(5x6)	BLDC_Motor_Driver_Components_Footprint:PG-TSON-8-3_INF
RT0603BRD07100KL	2	R1,R2	100K	Yes	C122538	Resistor	Resistor_SMD:R_0603_1608Metric
FRC0603F0000TS	10	R3,R6,R23,R24,R25,R26,R28,R29,R31,R33	0R	No,Yes	C2906974	Resistor	Resistor_SMD:R_0603_1608Metric
CRCW06039K09FKEA	1	R4	9.09K	Yes	C844591	Resistor	Resistor_SMD:R_0603_1608Metric
RC0603FR-0710KL,RT0603BRD0710KL	15	R5,R15,R17,R18,R19,R32,R34,R37,R43,R49,R52,R53,R54,R67,R68	10k	No,Yes	C95204,C98220	Resistor	Resistor_SMD:R_0603_1608Metric
FPM253WJR001TML	3	R7,R71,R74	1mR	Yes	C7419999	Resistor	Resistor_SMD:R_2512_6332Metric
25121WJ047JT4E	6	R8,R22,R72,R73,R75,R76	4R7	No	C25489	Resistor	Resistor_SMD:R_2512_6332Metric_Pad1.40x3.35mm_HandSolder
RC0603FR-0733RL	23	R11,R12,R16,R20,R21,R40,R41,R42,R44,R57,R60,R63,R64,R65,R66,R69,R70,R77,R78,R79,R80,R81,R82	33R	Yes	C108661	Resistor	Resistor_SMD:R_0603_1608Metric
RTT0360R0FTP	2	R13,R14	60R	Yes	C166918	Resistor	Capacitor_SMD:C_0603_1608Metric
RT0603BRD0718KL	5	R27,R30,R45,R47,R50	18K	Yes	C309650	Resistor	Resistor_SMD:R_0603_1608Metric
0603WAF3301T5E	2	R35,R36	3k3	Yes	C22978	Resistor	Resistor_SMD:R_0603_1608Metric
0603WAF5101T5E	2	R38,R39	5k1	Yes	C23186	Resistor	Resistor_SMD:R_0603_1608Metric
RT0603BRD0730KL	3	R46,R48,R51	30K	Yes	C723585	Resistor	Resistor_SMD:R_0603_1608Metric
FRC1206F1003TS	3	R55,R58,R61	100K	Yes	C2907368	Resistor	Resistor_SMD:R_1206_3216Metric
RT0603BRD072K7L	3	R56,R59,R62	2k7	Yes	C705760	Resistor	Resistor_SMD:R_0603_1608Metric
TL3301NF160QG	1	SW1	SW_Push	Yes	C273526	Push button switch, generic, two pins	Button_Switch_SMD:SW_Push_1P1T_NO_E-Switch_TL3301NxxxxxG

LM76005RNPR	1	U1	LM76005RNPR	Yes	C2866241	3.5-V to 60-V, 5-A synchronous step-down voltage converter	BLDC_Motor_Driver_Components_Footprint:QFN50P600X400X80-31N
TCAN334DR	1	U2	TCAN334DR		C2871143	3.3V CAN transceiver with CANFD (Flexible Data Rate)	BLDC_Motor_Driver_Components_Footprint:SOIC127P599X175-8N
TLV9061IDBVR	7	U3,U10,U11,U12,U13,U14,U15	TLV9061IDBVR	Yes	C398358	1.6mV Rail-to-Rail Input, Rail-to-Rail Output 0.5pA 103dB 538uA 10MHz 1 SOT-23-5 Amplifiers RoHS	BLDC_Motor_Driver_Components_Footprint:SOT95P280X145-5N
STM32G431RBT6	1	U4	STM32G431RBT6	Yes	C431633	Arm Cortex-M4 32b MCU+FPU,170MHz / 213DMIPS, up to 128 KB Flash, 32KB SRAM, rich analog, math accelerator	BLDC_Motor_Driver_Components_Footprint:LQFP64-10x10mm
LDL1117S33R	1	U5	LDL1117S33R	Yes	C435835	3.3V 18V Positive Fixed SOT-223-4 Voltage Regulators - Linear, Low Drop Out (LDO) Regulators RoHS	BLDC_Motor_Driver_Components_Footprint:SOT230P700X180-4N
DRV8353HRTAR	1	U6	DRV8353HRTAR	Yes	C2150766	Three Phase MOSFET 700mA 700mA WQFN-40-EP(6x6) Motor Drivers, Controllers RoHS	BLDC_Motor_Driver_Components_Footprint:RTA0040B
NCV4274ADT50RKG	1	U7	NCV4274ADT50RKG	Yes	C233416	5V 40V Positive Fixed TO-252-2(DPAK) Voltage Regulators - Linear, Low Drop Out (LDO) Regulators RoHS	BLDC_Motor_Driver_Components_Footprint:DPAK-3_ONS
25LC256-E/SN	1	U8	25LC256-E_SN	Yes	C1349834	2.5V~5.5V 256Kbit 10MHz SPI SOIC-8 Memory (ICs) RoHS	BLDC_Motor_Driver_Components_Footprint:SOIC127P599X175-8N
USBLC6-2SC6	1	U9	USBLC6-2SC6	Yes		Very low capacitance ESD protection diode, 2 data-line, SOT-23-6	Package_TO_SOT_SMD:SOT-23-6
LM74700QDRQ1	1	U16	LM74700QDRQ1	Yes	C33579657	20mV 11mA 65V 2.37A SOIC-8 OR Controllers, Ideal Diodes RoHS	BLDC_Motor_Driver_Components_Footprint:SOIC8_D_TEX
7M-24.000MAAJ-T	1	Y1	7M-24.000MAAJ-T	Yes	C2450233	Crystal 24MHz Â±30ppm 18pF 60Î© SMD3225-4P	BLDC_Motor_Driver_Components_Footprint:XTAL_7M-24.000MAAJ-T
ABS07-32.768KHZ-T	1	Y2	ABS07-32.768KHZ-T	Yes	C130253	Crystal 32.768kHz Â±20ppm 12.5pF 70kÎ© SMD3215-2P	BLDC_Motor_Driver_Components_Footprint:XTAL_ABS07-32.768KHZ-T
FRC1206J105 TS	1	R9	1M	Yes	C2907441	Resistor	Resistor_SMD:R_1206_3216Metric
TCC1206X7R103K101DT	1	C14	10nF	Yes	C377055	Unpolarized capacitor	Capacitor_SMD:C_1206_3216Metric